Capacity of Fading Broadcast Channels with Quality-of-Service Constraints

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***Abstract*— For fading broadcast channels, different capacity configurations such as ergodic, outage, minimum rate, or limited-jitter can be represented by a triplet of quality-of-service parameters from each user: maximum rate, minimum rate, and shortage probability. For each fading state, the channel capacity is obtained from evaluating a finite set of possible extreme points. Optimal power allocation strategy across fading states is shown to be water-filling with rates determined by the effective noise of the corresponding fading states, then combined with constant rate power allocation. Shortage capacity can be similarly obtained by removing the minimum rate constraints when the system is in one of the shortage fading states.**

I. Introduction

WHILE capacity for fading broadcast channels with different constraints are known: ergodic capacity [1], [2], outage capacity [3], and minimum rate capacity [4], this analysis focuses on the scenario where each user is able to specify a set of parameters that represents the quality-of-service (QoS) requirements on the channel. The user specifies a maximum rate needed by the applications, a minimum rate required, and a shortage probability over which the minimum rate stipulation might be waived. Effectively, when in shortage, the system resorts to a best-service resource provisioning scheme, in which minimum rates are no longer guaranteed. Similar to [3], there is also a system-wide parameter to specify whether common shortage mode, where all users declare shortage simultaneously, or independent shortage mode, where users declare shortage separately, is used.

This model is motivated by the desire to support heterogeneous applications with different QoS requirements over a common wireless channel. For example, delay-insensitive applications such as batch file transfers seek to maximize ergodic capacity with little QoS constraints. On the other hand, an adaptive audio/video streaming application would operate with a minimum rate needed for legible communication up to a maximum rate in which the source content is encoded. Any rate above this maximum does not provide additional utility. Whereas for wireless control applications, a fixed rate (therefore fixed delay) is much more desirable than raw capacity, hence the user would wish to set the minimum rate the same as the maximum rate to eliminate jitters.

The QoS requirement parameters are explained in Section II. Next, Section III presents the discrete-time memoryless broadcast channel system model used in the analysis. Section IV considers the scenario where the broadcast channel is static (i.e., no fading), while Section V extends the analysis results for a fading channel. Effects of shortage on capacity and power allocation is investigated in Section VI, and conclusions are presented in Section VII.

II. Quality-of-Service Constraints

Each user will specify the QoS requirements in a triplet of maximum rate R^m, minimum rate R^n, and shortage probability q. Let (R_j^m, R_j^n, q_j) be the QoS parameters given by user j, with $0 \leq R_j^n \leq R_j^m$ and $0 \leq q_j \leq 1$. Specifying a maximum rate means user j derives no additional utility beyond the rate R_j^m. The maximum rate can be interpreted as a hint that any capacity beyond R_j^m should be allocated to the other users.

On the other hand, setting a minimum rate with shortage probability means user j must receive rate equal to or above R_j^n with probability $1 - q_j$. With the remainder probability q_j, the system can declare shortage and the minimum rate is no longer guaranteed. Shortage capacity differs from outage capacity [3] in that when system is in shortage, while it is short of providing a minimum rate guarantee, transmission might still take place at a lower rate. Whereas when system is in outage, transmission is suspended completely. If transmission power is set to zero when system is in shortage, then shortage capacity reduces to outage capacity.

Besides the QoS requirements triplet (R_j^m, R_j^n, q_j) from each user, a system-wide parameter shortage mode is also to be specified. In common shortage mode, all users are required to declare shortage simultaneously, whereas independent shortage mode allows each user to have different shortage fading states. Naturally, if common shortage is specified, it is assumed that q_j's of all users are the same: $q_1 = \ldots = q_M = q$.

The QoS parameters can be set to represent a variety of channel capacity configurations. For example, having a minimum rate of zero and maximum rate of infinity leads to the ergodic capacity, while setting minimum rate the same as maximum rate represents zero-outage capacity. Different capacity configurations represented by the QoS parameters are shown in Table I.

Capacity Configuration	*QoS Parameters*
Ergodic [1], [2]	$(R^m, R^n, q) = (\infty, 0, 0)$
Zero-outage [3]	$R^m = R^n$, $q = 0$
Minimum rate [4]	$(R^m, R^n, q) = (\infty, r_{min}, 0)$
Outage [3]	$R^m = R^n$, $q = p_{out}$
Limited-jitter	$R^m - R^n = r_{jitter}$

TABLE I

Channel capacity configurations represented by QoS parameters.

III. System Model

The analysis adopts the same system model as defined in [2], [3], [4]. It assumes a discrete-time flat-fading Gaussian broadcast channel, with perfect channel state information at the transmitter and at all receivers. The transmitter varies its transmit power (thus controlling the transmission rate) under an average power constraint, and superposition coding with successive decoding [5] is used.

At each time instance i, the transmitter sends a signal $X_j(i)$ with power $p_j(i)$ to user j over bandwidth B. The average total transmission power over time cannot exceed $\overline{P}$. To characterize the channel, let v_j be the noise density of user j's channel, and $\sqrt{g_j(i)}$ its time-varying channel gain at time i. Then combine the channel gain and noise density to form the time-varying noise density $n_j(i) = v_j/g_j(i)$; since $n_j(i)$ incorporates the time-varying channel state, it will also be referred to as a fading state. Signals to all M users are sent simultaneously, hence the received signal for user j becomes

$$Y_j(i) = \sum_{k=1}^{M} X_k(i) + z_j(i) \tag{1}$$

where $z_j(i)$ is a Gaussian random variable with zero mean and variance $n_j(i)B$. It is assumed that the noise density vector $\mathbf{n}(i)$ = $(n_1(i)$,..., $n_M(i))$ are known to the transmitter and all receivers at each time instance i.

Using superposition coding with successive decoding, the rate for user j at fading state $\mathbf{n}$ is

$$R_j(\mathbf{n}) = B \log\left(1 + \frac{p_j(\mathbf{n})}{n_j B + \sum_{i=1}^{M} p_i(\mathbf{n})\, \mathbf{1}[n_j > n_i]}\right) \tag{2}$$

where $\mathbf{1}[\cdot]$ is the indicator function. Since B is simply a constant scaling factor, it can be assumed to be 1 without loss of generality. This analysis will focus on the case where the channel has two users ($M = 2$).

IV. Additive White Gaussian Noise Channel

First consider an additive white Gaussian noise (AWGN) channel where there is no fading. In this case, $n1$, $n2$ are constants with respect to time. Furthermore, assume $n_2 > n_1$ (user 2 has noisier channel); in the case of $n_1 > n_2$, the derivation remains valid with the subscripts 1, 2 reversed. Let user 1 and user 2 has QoS requirements $(R_1^m, R_1^n, 0)$ and $(R_2^m, R_2^n, 0)$, respectively (since channel has no fading, set shortage probability q_1 and q_2 to 0). To determine channel capacity, it is needed to maximize the achievable rates subject to the total power and QoS constraints:

$$\begin{aligned} &\max_{p_1,p_2}\ \alpha_1 \log\left(1 + \frac{p_1}{n_1}\right) + \alpha_2 \log\left(1 + \frac{p_2}{p_1 + n_2}\right) \\ &\quad \text{subject to: } p_1 + p_2 \le P,\ (R_1^m, R_1^n, 0),\ (R_2^m, R_2^n, 0) \end{aligned} \tag{3}$$

The scaling factor α_1 and α_2 allow capacity of user 1 and user 2 to be weighted differently; $0 < \alpha_1 < 1$, and $\alpha_2 = 1 - \alpha_1$. Note that from the definition $\alpha_1 \neq 0$ and $\alpha_2 \neq 0$; this is to simplify the boundary conditions in the maximization. However, since each factor can be arbitrarily close to zero, no generality is lost.

The channel capacity is a function of (p_1, p_2), the noise density vector $\mathbf{n} = (n_1, n_2)$, and the total available power P. The problem is to determine, for given P and $\mathbf{n}$, the optimal power allocation between the two users, i.e., solve for the optimal power allocation function

$$(p_1, p_2) = F_A(P, \mathbf{n}), \tag{4}$$

and the respective achieved capacity:

$$\begin{aligned} R_A(P, \mathbf{n}) &= \max_{p_1,p_2} R((p_1, p2), P, \mathbf{n}) && (5)\\ &= R(F_A(P, \mathbf{n}), P, \mathbf{n}). && (6) \end{aligned}$$

Successive decoding allows user 1 to subtract out the signal power of user 2. Therefore, user 1 is not affected by user 2. To user 2, however, user 1's signal power p_1 is effectively noise. For user 2 to maintain a constant rate R_2^m, as derived in [4], its power p_2 needs to be accordingly increased whenever p_1 is increased:

$$p_2 = (e^{R_2^m} - 1)p_1 + n_2(e^{R_2^m} - 1). \tag{7}$$

Likewise, if a constant rate of R_2^n is to be maintained:

$$p_2 = (e^{R_2^n} - 1)p_1 + n_2(e^{R_2^n} - 1). \tag{8}$$

For convenience, define the following constants:

$$\begin{aligned} P_1^m &\triangleq n_1(e^{R_1^m} - 1) \\ P_1^n &\triangleq n_1(e^{R_1^n} - 1) \\ G_m &\triangleq e^{R_2^m} - 1 \\ G_n &\triangleq e^{R_2^n} - 1 \\ P_2^m &\triangleq G_m(P_1^m + n_2) \\ P_2^n &\triangleq G_n(P_1^n + n_2). \end{aligned}$$

Then the constraints of (3) can be expressed in terms of the following five inequalities:

$$\begin{aligned} p_1 + p_2 &\le P && (9)\\ p_1 &\le P_1^m && (10)\\ p_1 &\ge P_1^n && (11)\\ p_2 &\le G_m(p_1 + n_2) && (12)\\ p_2 &\ge G_n(p_1 + n_2) && (13) \end{aligned}$$

Note that all of the constraints are linear in p_1, p_2. Let Ω denote the region over p_1, p_2 where all constraints are met. Since $P \ge 0$ and $0 \le R_j^n \le R_j^m$ for each user $j = 1, 2$, it implies $P_1^m \ge P_1^n$, $G_m \ge G_n$ and $P_2^m \ge P_2^n$. These conditions lead to a contiguous and closed region Ω on the p_1-p_2 plane, as shown in Fig. 1. Let the lines ℓ_T, ℓ_{M_1}, ℓ_{N_1}, ℓ_{M_2} and ℓ_{N_2} denote the boundaries of Ω, which correspond, respectively, to constraints (9)–(13) with equality being taken, as summarized in Table II below.

An extreme point is defined by the intersection of two boundaries. Note that a point (p_1, p_2) can be represented as a vector $\mathbf{p} = p_1\, \mathbf{i} + p_2\, \mathbf{j}$, where $\mathbf{i}$, $\mathbf{j}$ are unit vectors in the p_1, p_2 directions, respectively. Six extreme points $\mathbf{M_1}$, $\mathbf{N_1}$, $\mathbf{M_2}$, $\mathbf{N_2}$, $\mathbf{X}$ and $\mathbf{Y}$ are of particular interest; their definitions are tabulated in Table III.

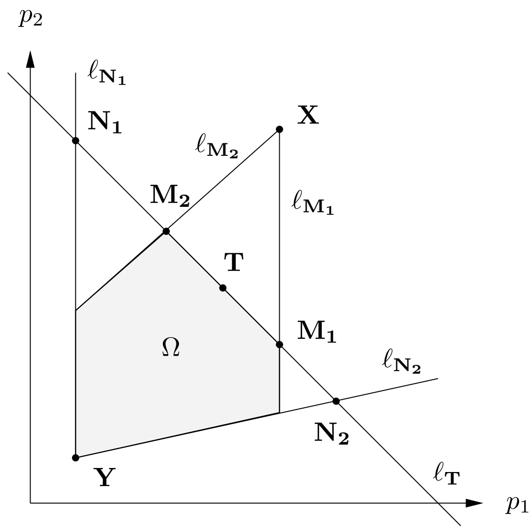

Fig. 1. QoS constraints on power allocation for user 1, 2.

Line	*Constraint*	*Constraint Type*
ℓ_T	(9)	total average power
ℓ_{M_1}	(10)	maximum rate for user 1
ℓ_{N_1}	(11)	minimum rate for user 1
ℓ_{M_2}	(12)	maximum rate for user 2
ℓ_{N_2}	(13)	minimum rate for user 2

TABLE II

CONSTRAINT BOUNDARIES.

Point	*Intersection*	*Constraint Type*
$\mathbf{M_1}$	ℓ_T, ℓ_{M_1}	maximum rate for user 1
$\mathbf{N_1}$	ℓ_T, ℓ_{N_1}	minimum rate for user 1
$\mathbf{M_2}$	ℓ_T, ℓ_{M_2}	maximum rate for user 2
$\mathbf{N_2}$	ℓ_T, ℓ_{N_2}	minimum rate for user 2
$\mathbf{X}$	ℓ_{M_1}, ℓ_{M_2}	minimum rate for both users
$\mathbf{Y}$	ℓ_{N_1}, ℓ_{N_2}	minimum rate for both users

TABLE III

CONSTRAINT BOUNDARY EXTREME POINTS.

The channel capacity function

$$R(\mathbf{p}) = \alpha_1 \log\left(1 + \frac{p_1}{n_1}\right) + \alpha_2 \log\left(1 + \frac{p_2}{p_1 + n_2}\right), \quad (14)$$

being differentiable everywhere, is differentiable on the closed region Ω. Therefore, $R(\mathbf{p})$ must take on an absolute maximum on Ω, which can be found as follows:

1) First calculate the the gradient of $R(\mathbf{p})$:

$$\nabla R(\mathbf{p}) = \frac{\partial R}{\partial p_1}(\mathbf{p})\,\mathbf{i} + \frac{\partial R}{\partial p_2}(\mathbf{p})\,\mathbf{j} \quad (15)$$

$$= \left(\frac{\alpha_1}{p_1 + n_1} - \frac{\alpha_2 p_2}{(p_1 + n_2)(p_1 + p_2 + n_2)}\right)\mathbf{i} + \frac{\alpha_2}{p_1 + p_2 + n_2}\,\mathbf{j}. \quad (16)$$

The gradient, defined everywhere, is never $\mathbf{0}$. In particular, since $\alpha_2 > 0$, its $\mathbf{j}$-component will always be positive. Therefore, the maximum point of $R(\mathbf{p})$ is on one of the boundaries of Ω.

2) Consider the boundaries ℓ_{M_2} and ℓ_{N_2}. Define unit vectors $\mathbf{u_m}$, $\mathbf{u_n}$, respectively, to have the same directions as ℓ_{M_2}, ℓ_{N_2} (in the increasing $p1$-$p2$ polarity):

$$\mathbf{u_m} = \frac{\mathbf{i} + G_m\,\mathbf{j}}{\sqrt{(G_m)^2 + 1}}, \quad (17)$$

$$\mathbf{u_n} = \frac{\mathbf{i} + G_n\,\mathbf{j}}{\sqrt{(G_n)^2 + 1}}. \quad (18)$$

The directional derivatives of $R(\mathbf{p})$ with respective to $\mathbf{u_m}$ and $\mathbf{u_n}$, respectively, are given by the dot products:

$$R'_{\mathbf{u_m}}(\mathbf{p}) = \nabla R(\mathbf{p}) \cdot \mathbf{u_m} \quad (19)$$

$$= \frac{1}{\sqrt{(G_m)^2 + 1}} \frac{\alpha_1}{p_1 + n_2}, \quad (20)$$

$$R'_{\mathbf{u_n}}(\mathbf{p}) = \nabla R(\mathbf{p}) \cdot \mathbf{u_n} \quad (21)$$

$$= \frac{1}{\sqrt{(G_n)^2 + 1}} \frac{\alpha_1}{p_1 + n_2}. \quad (22)$$

Since both $R'_{\mathbf{u_m}}(\mathbf{p})$ and $R'_{\mathbf{u_n}}(\mathbf{p})$ are always positive, $R(\mathbf{p})$ is monotonically increasing in the directions of $\mathbf{u_m}$ and $\mathbf{u_n}$. Therefore, if the maximum point lies on ℓ_{M_2} or ℓ_{N_2}, it has to be one of the extreme points $\mathbf{M_2}$, $\mathbf{N_2}$, or $\mathbf{X}$.

3) Next consider the boundaries ℓ_{M_1} and ℓ_{N_1}. Both of them are parallel to the p_2 axis, hence the derivative of $R(\mathbf{p})$ in the direction of $\mathbf{j}$ is

$$R'_{\mathbf{j}}(\mathbf{p}) = \nabla R(\mathbf{p}) \cdot \mathbf{j} \quad (23)$$

$$= \frac{\alpha_2}{p_1 + p_2 + n_2}. \quad (24)$$

Similarly, since the directional derivative $R'_{\mathbf{j}}(\mathbf{p})$ is always positive, $R(\mathbf{p})$ is monotonically increasing in the direction of $\mathbf{j}$. Therefore, if the maximum point lies on ℓ_{M_1} or ℓ_{N_1}, it has to be one of the extreme points $\mathbf{M_1}$ or $\mathbf{N_1}$.

4) If the maximum point does not lie on ℓ_{M_2}, ℓ_{N_2}, ℓ_{M_1}, or ℓ_{N_1}, then it has to be on the boundary ℓ_T within the open interval between points $\mathbf{M_1}$ and $\mathbf{M_2}$. Denote such point as $\mathbf{T}$. Then optimization (3) reduces to a standard one-dimensional maximization problem. The derivative of the channel capacity along ℓ_T with respect to p_1 is

$$\frac{dR_{\ell_T}}{dp_1}(p_1) = \frac{(\alpha_1 - \alpha_2)p_1 + \alpha_1 n_2 - \alpha_2 n_1}{(p_1 + n_1)(p_1 + n_2)}. \quad (25)$$

Since $n_2 > n_1$, the derivative is always positive for $\alpha_1 \geq \alpha_2$. In such case $R_{\ell_T}(p_1)$ is monotonically increasing in the direction of p_1, so $\mathbf{M_1}$ or $\mathbf{N_2}$ are the possible maximum points rather than $\mathbf{T}$. For $\alpha_1 < \alpha_2$, $\mathbf{T}$ is given by

$$\mathbf{T} = \left(\frac{\alpha_1 n_2 - \alpha_2 n_1}{\alpha_2 - \alpha_1}, P - \frac{\alpha_1 n_2 - \alpha_2 n_1}{\alpha_2 - \alpha_1}\right). \quad (26)$$

5) Finally, when available total power equals the minimum power requirement $P = P_1^n + P_2^n$. All of the above operating points close in to meet at the minimum rate extreme point $\mathbf{Y}$.

Therefore, there are only seven potential maximum point candidates: $\mathbf{M_1}$, $\mathbf{N_1}$, $\mathbf{M_2}$, $\mathbf{N_1}$, $\mathbf{T}$, $\mathbf{X}$ and $\mathbf{Y}$. Thus the maximum of $R(\mathbf{p})$ can be obtained by evaluating the candidate points that lie on the boundaries of Ω, and choose the one that yields the largest value, i.e.,

$$F_A(P,\mathbf{n}) = \mathbf{v} \text{ such that } R(\mathbf{v}) \geq R(\mathbf{v}') \; \forall \; \mathbf{v}' \neq \mathbf{v}, \\ \text{where } \mathbf{v},\mathbf{v}' \in \{\mathbf{M_1},\mathbf{N_1},\mathbf{M_2},\mathbf{N_2},\mathbf{T},\mathbf{X},\mathbf{Y}\} \cap \Omega. \tag{27}$$

The coordinates of the possible maximum points are presented in Table IV. To evaluate the channel capacity at the possible maximum points, substitute the coordinates into (14) to obtain:

$$R(\mathbf{M_1}) = \alpha_2 \log\left(1 + \frac{P - P_1^m}{P_1^m + n_2}\right) + \alpha_1 R_1^m \tag{28}$$

$$R(\mathbf{N_1}) = \alpha_2 \log\left(1 + \frac{P - P_1^n}{P_1^n + n_2}\right) + \alpha_1 R_1^n \tag{29}$$

$$R(\mathbf{M_2}) = \alpha_1 \log\left(1 + \frac{P - G_m n_2}{(G_m+1)n_1}\right) + \alpha_2 R_2^m \tag{30}$$

$$R(\mathbf{N_2}) = \alpha_1 \log\left(1 + \frac{P - G_n n_2}{(G_n+1)n_1}\right) + \alpha_2 R_2^n \tag{31}$$

$$R(\mathbf{T}) = \alpha 2 \log\left(1 + \frac{P - \frac{\alpha_2 n_1 - \alpha_1 n_2}{\alpha_1 - \alpha_2}}{\frac{\alpha_2}{\alpha_1 - \alpha_2}(n_1 - n_2)}\right) \\ + \alpha_1 \log\left(1 + \frac{\alpha_2 n_1 - \alpha_1 n_2}{(\alpha_1 - \alpha_2) n_1}\right) \tag{32}$$

$$R(\mathbf{X}) = \alpha_1 R_1^m + \alpha_2 R_2^m \tag{33}$$

$$R(\mathbf{Y}) = \alpha_1 R_1^n + \alpha_2 R_2^n \tag{34}$$

Then the channel capacity associated with the optimal power allocation between the users is

$$R_A(P,\mathbf{n}) = R(F_A(P,\mathbf{n}),P,\mathbf{n}), \tag{35}$$

which is given by one of the equations in (28)–(34).

V. Fading Channel

In this section, zero-shortage capacity of a fading channel will be studied. Effects of shortage are analyzed in Section VI. For a given total power $P(\mathbf{n})$ and noise density $\mathbf{n}$, optimally allocating the power between the users is solved in Section IV; the power allocation between user 1 and user 2 is given by (27) and the maximum rate achieved is either the constant rates (33), (34), or as given by (37). Hence, the remaining step is to determine the optimal power allocation across fading states $\mathbf{n}$.

In a fading channel, n_1, n_2 are random variables with known joint probability distribution. Consider based on the noise density vector $\mathbf{n} = (n_1, n_2)$, the total power $P(\mathbf{n})$ can be varied. Partition the noise density vector into states $\{S_{\mathbf{M_1}}$, $S_{\mathbf{N_1}}$, $S_{\mathbf{M_2}}$, $S_{\mathbf{N_2}}$, $S_{\mathbf{T}}$, $S_{\mathbf{X}}$, $S_{\mathbf{Y}}\}$, where $S_{\mathbf{v}}$ corresponds to the values of $\mathbf{n}$ that results in the channel capacity taking on maximum at $\mathbf{v}$ for a given total power P, i.e.,

$$\mathbf{n} \in S_{\mathbf{v}} \text{ if } F_A(P,\mathbf{n}) = \mathbf{v}, \\ \text{where } \mathbf{v} \in \{\mathbf{M_1},\mathbf{N_1},\mathbf{M_2},\mathbf{N_2},\mathbf{T},\mathbf{X},\mathbf{Y}\}. \tag{36}$$

Therefore, the channel capacity for a given noise density state $S_{\mathbf{v}}$ and total power P is given by (28)–(34). For states $S_{\mathbf{X}}$ and $S_{\mathbf{Y}}$, the channel has constant rates. For the other states, the channel capacity has the following form:

$$R_A(P(\mathbf{n}),\mathbf{n}) = A_{\mathbf{v}} \log\left(1 + \frac{P(\mathbf{n}) - B_{\mathbf{v}}}{C_{\mathbf{v}}}\right) + D_{\mathbf{v}} \tag{37}$$

State	$A_{\mathbf{v}}$	$n^*_{\mathbf{v}}(n_1,n_2)$
$S_{\mathbf{M_1}}$	α_2	n_2
$S_{\mathbf{N_1}}$	α_2	n_2
$S_{\mathbf{M_2}}$	α_1	$(G_m+1)n_1 - G_m n_2$
$S_{\mathbf{N_2}}$	α_1	$(G_n+1)n_1 - G_n n_2$
$S_{\mathbf{T}}$	α_2	n_2

TABLE VI

Water-filling parameters for each noise density state.

where $A_{\mathbf{v}}$, $B_{\mathbf{v}}$, $C_{\mathbf{v}}$ and $D_{\mathbf{v}}$ are parameters specific to the noise density state $S_{\mathbf{v}}$. The values of these parameters are tabulated in Table V.

To find the optimal power allocation strategy for each $\mathbf{n}$, i.e., to determine $P(\mathbf{n})$, it is needed to maximize the following:

$$\max_{P(\mathbf{n})} \; \mathrm{E}_{\mathbf{n}}\left[R((p_1(\mathbf{n}),p_2(\mathbf{n})),P(\mathbf{n}),\mathbf{n})\right] \tag{38}$$

$$= \max_{P(\mathbf{n})} \\ \mathrm{E}_{\mathbf{n}}\left[\alpha_1 \log\left(1 + \frac{p_1(\mathbf{n})}{n_1}\right) + \alpha_2 \log\left(1 + \frac{p_2(\mathbf{n})}{p_1(\mathbf{n}) + n_2}\right)\right] \\ \text{subject to: } \mathrm{E}_{\mathbf{n}}[P(\mathbf{n})] \leq \overline{P}, \tag{39}$$

where $p_1(\mathbf{n}) + p_2(\mathbf{n}) \leq P(\mathbf{n})$.

For $S_{\mathbf{X}}$ and $S_{\mathbf{Y}}$, power allocation is dictated by the maximum and minimum rates:

$$P_{\mathbf{X}}(\mathbf{n}) = P_1^m + P_2^m \tag{40}$$

$$P_{\mathbf{Y}}(\mathbf{n}) = P_1^n + P_2^n \tag{41}$$

where $P_{\mathbf{X}}(\mathbf{n})$, $P_{\mathbf{Y}}(\mathbf{n})$ are the power allocation strategy for states $S_{\mathbf{X}}$, $S_{\mathbf{Y}}$, respectively. For the other states, substitute (37) in (38) to obtain:

$$\max_{P(\mathbf{n})} \; \mathrm{E}_{\mathbf{n}}\left[A_{\mathbf{v}} \log\left(1 + \frac{P(\mathbf{n}) - B_{\mathbf{v}}}{C_{\mathbf{v}}}\right) + D_{\mathbf{v}}\right] \\ \text{subject to: } \mathrm{E}_{\mathbf{n}}[P(\mathbf{n})] \leq \overline{P}. \tag{42}$$

Using Lagrangian method, form

$$J(P(\mathbf{n})) = A_{\mathbf{v}} \log\left(1 + \frac{P(\mathbf{n}) - B_{\mathbf{v}}}{C_{\mathbf{v}}} + D_{\mathbf{v}}\right) \\ - \lambda(\mathrm{E}_{\mathbf{n}}[P(\mathbf{n})] - \overline{P}), \tag{43}$$

then set $\frac{\partial J}{\partial P(\mathbf{n})}$ to zero to obtain:

$$P(\mathbf{n}) = \begin{cases} A_{\mathbf{v}}\frac{1}{\lambda} - (C_{\mathbf{v}} - B_{\mathbf{v}}) & A_{\mathbf{v}}\frac{1}{\lambda} \geq C_{\mathbf{v}} - B_{\mathbf{v}} \\ 0 & \text{else} \end{cases} \tag{44}$$

where $\frac{1}{\lambda}$ is the water-filling level. Define effective noise $n^*_{\mathbf{v}}(n_1,n_2) = C_{\mathbf{v}} - B_{\mathbf{v}}$. The values of $A_{\mathbf{v}}$ and $n^*_{\mathbf{v}}$ for each state are summarized in Table VI. Notice that states S_{M_1}, S_{N_2} and S_T all have $A_{\mathbf{v}} = \alpha_2$ and the same effective noise $n^*_{\mathbf{v}} = n_2$. Therefore, there are only three water-filling equations:

$$P_{\mathbf{M_1},\mathbf{N_1},\mathbf{T}}(\mathbf{n}) = \lfloor \alpha_2 \frac{1}{\lambda} - n_2 \rfloor_+ \tag{45}$$

$$P_{\mathbf{M_2}}(\mathbf{n}) = \lfloor \alpha_1 \frac{1}{\lambda} - (G_m+1)n_1 + G_m n_2 \rfloor_+ \tag{46}$$

$$P_{N_2}(\mathbf{n}) = \lfloor \alpha_1 \frac{1}{\lambda} - (G_n+1)n_1 + G_n n_2 \rfloor_+ \tag{47}$$

Next, it is needed to determine which power allocation equation to use for a given fading state $\mathbf{n}$. Consider the power

Point	p_1	p_2	*Constraint Type*
$\mathbf{M_1}$	P_1^m	$P - P_1^m$	maximum rate for user 1
$\mathbf{N_1}$	P_1^n	$P - P_1^n$	minimum. rate for user 1
$\mathbf{M_2}$	$\frac{1}{G_m+1}(P - G_m n_2)$	$\frac{G_m}{G_m+1}(P + n_2)$	maximum rate for user 2
$\mathbf{N_2}$	$\frac{1}{G_n+1}(P - G_n n_2)$	$\frac{G_n}{G_n+1}(P + n_2)$	minimum rate for user 2
$\mathbf{T}$	$\frac{\alpha_1 n_2 - \alpha_2 n_1}{\alpha_2 - \alpha_1}$	$P - \frac{\alpha_2 n_1 - \alpha_1 n_2}{\alpha_1 - \alpha_2}$	total average power
$\mathbf{X}$	P_1^m	P_2^m	maximum rate for both users
$\mathbf{Y}$	P_1^n	P_2^n	minimum rate for both users

TABLE IV

POSSIBLE MAXIMUM POINTS.

State	$A_{\mathbf{v}}$	$B_{\mathbf{v}}$	$C_{\mathbf{v}}$	$D_{\mathbf{v}}$
$S_{\mathbf{M_1}}$	α_2	P_1^m	$P_1^m + n_2$	$\alpha_1 R_1^m$
$S_{\mathbf{N_1}}$	α_2	P_1^n	$P_1^n + n_2$	$\alpha_1 R_1^n$
$S_{\mathbf{M_2}}$	α_1	$G_m n_2$	$(G_m + 1)n_1$	$\alpha_2 R_2^m$
$S_{\mathbf{N_2}}$	α_1	$G_n n_2$	$(G_n + 1)n_1$	$\alpha_2 R_2^n$
$S_{\mathbf{T}}$	α_2	$\frac{\alpha_2 n_1 - \alpha_1 n_2}{\alpha_1 - \alpha_2}$	$\frac{\alpha_2}{\alpha_1 - \alpha_2}(n_1 - n_2)$	$\alpha_1 \log\left(1 + \frac{\alpha_2 n_1 - \alpha_1 n_2}{(\alpha_1 - \alpha_2) n_1}\right)$

TABLE V

RATE FUNCTION PARAMETERS FOR EACH NOISE DENSITY STATE.

allocation $P_{\mathbf{v}}(\mathbf{n})$ in the direction n_1 for each state. Effective noise of states $S_{\mathbf{M_1}}$, $S_{\mathbf{N_1}}$, $S_{\mathbf{T}}$ does not depend on n_1, so allocated power is simply the constant $\alpha_2 \frac{1}{\lambda} - n_2$, whereas states $S_{\mathbf{M_2}}$ and $S_{\mathbf{N_2}}$ have linear power allocation with slopes, respectively, $-(G_m+1)$ and $-(G_n+1)$, as illustrated in Fig. 2.

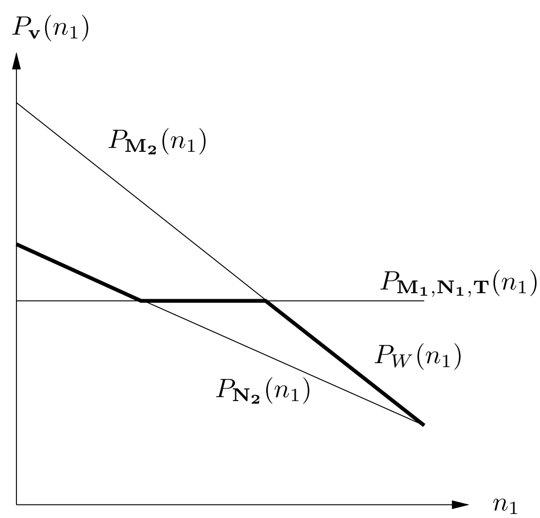

Fig. 2. Power allocation for each noise density state $S_{\mathbf{v}}$ in the direction of n_1, with the combined water-filling power allocation $P_W(n_1)$ shown in bold.

Observe that as given in Table IV, as n_1 increases, T moves from $\mathbf{N_2}$ towards $\mathbf{M_2}$; therefore, in the direction of increasing n_1 the states can only appear in this order: $\{S_{\mathbf{N_2}}, S_{\mathbf{T}}, S_{\mathbf{M_2}}\}$. Coupled with the fact that slope of $P_{\mathbf{T}}(n_1) \geq$ slope of $P_{\mathbf{N_2}}(n_1) \geq$ slope of $P_{\mathbf{M_2}}(n_1)$ with respect to n_1, the water-filling power allocation can be represented compactly by

$$P_W(\mathbf{n}) = \min\{\max\{P_{\mathbf{N_2}}(\mathbf{n}), P_{\mathbf{T}}(\mathbf{n})\}, P_{\mathbf{M_2}}(\mathbf{n})\}. \quad (48)$$

Alternatively, in the direction of n_2, increasing n_2 while keeping n_1 fixed implies the state occurrence order $\{S_{\mathbf{M_2}}, S_{\mathbf{N_1}}\}$ and $\{S_{\mathbf{M_1}}, S_{\mathbf{N_2}}\}$. With slope of $P_{\mathbf{N_1}}(n_2) >$ slope of $P_{\mathbf{M_2}}(n_2)$, and slope of $P_{\mathbf{M_1}}(n_2) >$ slope of $P_{\mathbf{N_2}}(n_2)$, it leads to the same conclusion. Finally, combine with the maximum and minimum rate constraints, the optimal power allocation strategy is:

$$P^*(\mathbf{n}) = \min\{\max\{P_W(\mathbf{n}), P_{\mathbf{Y}}(\mathbf{n})\}, P_{\mathbf{X}}(\mathbf{n})\}. \quad (49)$$

The water-filling level $\frac{1}{\lambda}$ can be determined by choosing the maximum water-filling level allowed by the average total power constraint:

$$\lambda = \lambda^* \text{ such that } \max_{\lambda^*} \; \mathrm{E}_{\mathbf{n}}[P(\mathbf{n}, \lambda^*)] \leq \overline{P}, \quad (50)$$

and the channel capacity achieved is simply the weighted average of the capacity over the fading states:

$$R^* = \mathrm{E}_{\mathbf{n}}[R(P^*(\mathbf{n}))]. \quad (51)$$

This is illustrated in Fig. 3, where the dashed line denotes a lower water-filling level than the one in bold.

VI. SHORTAGE PROBABILITY

A. Common Shortage

In the common shortage mode, when the system declares shortage for a certain fading state $\mathbf{n}$, the minimum rate constraints for all M users are removed. Naturally, all users have the same shortage probability q. Effectively, this is same optimization problem as before with minimum rates $R_j^n = 0 \; \forall j = 1, \ldots, M$. Therefore, similar procedures from Section IV can be used to deduce the optimal power allocation.

1) First calculate the zero-shortage power allocation function as previously derived. Write $P^*(\mathbf{n}, \lambda)$ to denote power allocation in terms of its water-filling level λ,

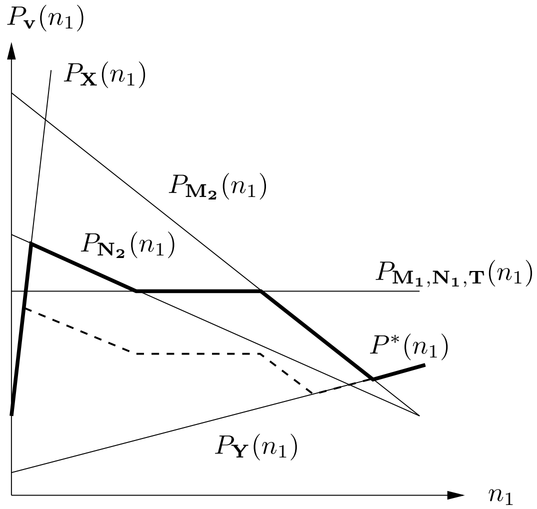

Fig. 3. Water-filling combined with constant rate power allocation. The dashed line denotes a lower water-filling level than the one in bold.

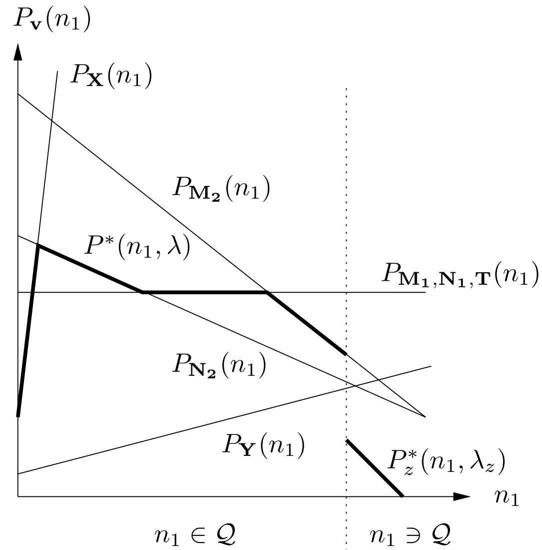

Fig. 4. Power allocation with shortage probability.

and $P^*(\mathbf{n})$ when λ is determined by the average total power constraint (50).

2) Repeat the same calculations with the minimum rate constraints removed, i.e., set $R_j^n = 0\ \forall j$; denote such power allocation function obtained as $P_z^*(\mathbf{n}, \lambda_z)$, and when the water-filling level λ_z is determined, as $P_z^*(\mathbf{n})$.
3) Select the set of shortage fading states $\mathcal{Q}$ to contain the fading states $\mathbf{n}$ that results in maximum power savings while not exceeding the specified shortage probability q:

$$\max_{\mathcal{Q}} \sum_{\mathbf{n}\in\mathcal{Q}} P^*(\mathbf{n}) - P_z^*(\mathbf{n}) \text{ such that } \sum_{\mathbf{n}\in\mathcal{Q}} \Pr(\mathbf{n}) \leq q \tag{52}$$

4) As shown in Fig. 4, the final shortage capacity power allocation strategy $P_{\mathcal{Q}}(\mathbf{n})$ is obtained from combining $P^*(\mathbf{n}, \lambda)$ and $P_z^*(\mathbf{n}, \lambda_z)$ over their respective associated domain of $\mathbf{n}$.

$$P_{\mathcal{Q}}^*(\mathbf{n}, \lambda, \lambda_z) = \begin{cases} P^*(\mathbf{n}, \lambda) & \mathbf{n} \ni \mathcal{Q}, \\ P_z^*(\mathbf{n}, \lambda_z) & \mathbf{n} \in \mathcal{Q}. \end{cases} \tag{53}$$

The water-filling levels λ, λ_z need to be recalculated to satisfy the average total power constraint (50). However, since the removal of minimum rate constraints represents a discontinuous change of channel capacity with respective to $\mathbf{n}$, the water-filling levels λ and λ_z are independent. After constraint (50) is applied, there still is an extra degree of freedom over division of power between λ and λ_z. Unfortunately, as λ and λ_z have no closed form analytic expressions, standard optimization techniques cannot be applied. Notwithstanding, in practice when the shortage probability q specified is small, λ_z is often zero. This is because small q limits $\mathcal{Q}$ to only contain fading states where noise density $\mathbf{n}$ is large, over which the optimal power allocation strategy simply is to suspend transmission. In this case shortage capacity reduces to outage capacity.

B. Independent Shortage

In independent shortage mode, each user specifies a shortage probability $q_j, j = 1, \ldots, M$, over which the minimum rate constraint for that user is not guaranteed, i.e., $R_j^n = 0$. Calculating the optimal power allocation for independent shortage capacity is similar to that of common shortage. However, it is needed to compare the zero-shortage power allocation function to that of each of the $2^M - 1$ permutations of different users in shortage. Moreover, it is needed to divide the power among the $2^M - 1$ water-filling levels. As described in Section VI-A, as there is no closed form expression for the channel capacity, standard optimization techniques cannot be readily applied. Further research work is needed to devise the optimal power allocation strategy for this scenario.

VII. Conclusion

In this analysis it has been shown that a variety of channel capacity configurations, e.g., ergodic, outage, minimum rate, limited-jitter, or a heterogeneous combination of them, can be represented by a triplet of QoS parameters from each user. Optimal power allocation between the users can be readily determined by evaluating a finite set of possible extreme points. The resulting channel capacity with respect to total power is either constant, or has a common form that is logarithmically proportional to the total power relative to an effective noise. This common capacity expression allows power allocation across fading states to be optimized, which is shown to be water-filling with rates determined by the effective noise of the corresponding fading states, then combined with constant rate power allocation. Finally, it is shown that shortage capacity can be similarly obtained by removing the minimum rate constraints when the system is in one of the shortage fading states. Therefore, the QoS parameter model provides a unifying framework in which channel capacity configurations with heterogeneous requirements from different users can be analyzed to obtain the respective channel capacity and optimal power allocation strategy.

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